Recall the Heat Eqn

Thermal Diffusity

$$\underbrace{\frac{\partial u}{\partial t}}_{\text{Temp change}} = \alpha \, \nabla^2 u$$

Temp change

Net local curvature
[Hot ⇒ -ve
Cold ⇒ +ve]

At steady state, $\frac{\partial u}{\partial t} = 0$,

$$0 = \alpha \cdot \nabla^2 u$$

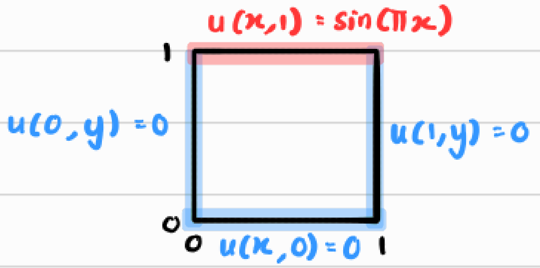
$$\boxed{\nabla^2 u = 0}$$ [since $\alpha \neq 0$]

This is the Laplace's Eqn .

Example 1

$\frac{\partial u}{\partial t} = 0 \Rightarrow \nabla^2 u = 0$

Let there be a thin, flat, square metal plate in steady state. Find $u(x,y)$



Domain

$0 \le x \le 1$
$0 \le y \le 1$

Original PDE : $\nabla^2 u = 0$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad — \circledast$$

Assume $u(x,y) = X(x) \cdot Y(y)$

$$\Rightarrow \begin{cases} \frac{\partial^2 u}{\partial x^2} = X''(x) \cdot Y(y) \\ \frac{\partial^2 u}{\partial y^2} = X(x) \cdot Y''(y) \end{cases} \quad — ①$$

① → ⊛

$$X''Y + XY'' = 0$$

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$$

Separation constant

$\therefore X'' + \lambda X = 0 \quad — ②$

$Y'' - \lambda Y = 0 \quad — ③$

Condition Interpretation

$u(0,y) = 0$

$X(0) \cdot Y(y) = 0$

$\boxed{X(0) = 0}$, $Y(y) = 0$ [Reject ∴ Trivial]

$u(x,0) = 0$

$X(x) \cdot Y(0) = 0$

$X(x) = 0$ [Reject ∴ Trivial], $\boxed{Y(0) = 0}$

$u(1,y) = 0$

$X(1) \cdot Y(y) = 0$

$\boxed{X(1) = 0}$, $Y(y) = 0$ [Rejected ∴ Trivial]

Solving for ②

Characteristic Eqn

$\Rightarrow r^2 + \lambda = 0$

$r = \pm\sqrt{-\lambda}$

$x=0, x=1$

Since $X(x)$ have 2 roots, $\lambda > 0$

Let $\lambda = w^2$

General soln : $X(x) = C_1 \cos(wx) + C_2 \sin(wx)$

Since $X(0) = 0 \Rightarrow C_1 = 0$

$\therefore X(x) = C_2 \sin(wx)$

Since $X(1) = 0 \Rightarrow \sin(wx) = 0$

$w = n\pi$ for $n = 1, 2, 3, \ldots$

$\therefore X_n(x) = \sin(n\pi x)$

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Solving for ③

Since $w = n\pi \Rightarrow \lambda = (n\pi)^2$

$\therefore \; Y'' - (n\pi)^2 Y = 0$

Characteristic Eqn : $r^2 = (n\pi)^2$

$r = \pm n\pi$

$\therefore$ General soln : $Y(y) = A_n \cosh(n\pi y) + B_n \sinh(n\pi y)$

Since $Y(0) = 0 \Rightarrow A_n = 0$

$\therefore \; Y_n(y) = B_n \sinh(n\pi y)$

$$\therefore \; u_n(x,y) = X_n(x) \cdot Y_n(y)$$
$$= B_n \sin(n\pi x) \sinh(n\pi y)$$

$$u(x,y) = \sum_{n=1}^{\infty} B_n \sin(n\pi x) \sinh(n\pi y)$$

Since $u(x,1) = \sin(\pi x)$

$$\sum_{n=1}^{\infty} B_n \sin(n\pi x) \sinh(n\pi) = \sin(\pi x)$$

By comparison $\Rightarrow B_1 \sin(\pi x) \sinh(\pi) = \sin(\pi x)$

$$B_1 = \frac{1}{\sinh(\pi)}$$

$$\therefore \; u(x,y) = \frac{1}{\sinh(\pi)} \sin(\pi x) \sinh(\pi y) \;\#$$

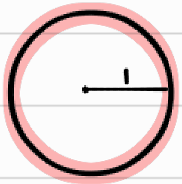
Laplace on disc

Laplacian [Polar Form] : $\boxed{\nabla^2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}}$

$\frac{\partial u}{\partial t} = 0 \Rightarrow \nabla^2 u = 0$

Example 1

Let there be a flat, circular disc $\bar{w}$ radius 1 in steady state. Find $u(r,\theta)$



$u(1,\theta) = \sin\theta$

Domain

$0 \le \theta \le 2\pi$

$0 \le r \le 1$

Original PDE : $\nabla^2 u = 0$

$$\Rightarrow \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0 \quad \text{—} \; (*)$$

Assume $u(r,\theta) = R(r) \cdot T(\theta)$

$$\Rightarrow \left\{ \begin{array}{l} \frac{\partial u}{\partial r} = R'(r) \cdot T(\theta) \\ \frac{\partial^2 u}{\partial r^2} = R''(r) \cdot T(\theta) \\ \frac{\partial^2 u}{\partial \theta^2} = R(r) \cdot T''(\theta) \end{array} \right. \quad \text{—} \; (1)$$

$(1) \to (*)$

$$R''T + \frac{1}{r}R'T + \frac{1}{r^2}RT'' = 0$$

$$r^2R''T + rR'T + RT'' = 0$$

$$\frac{r^2R''}{R} + \frac{rR'}{R} = -\frac{T''}{T} = \lambda$$

Separation constant

$$\therefore r^2R'' + rR' - \lambda R = 0 \quad \text{—} \; (2)$$

$$T'' + \lambda T = 0 \quad \text{—} \; (3)$$

Solving for (3)

Characteristic Eqn : $r^2 + \lambda = 0$

$$r = \pm\sqrt{-\lambda}$$

Since $T(\theta) = T(\theta + 2\pi)$ [aka. periodic]

Case 1 : $\lambda > 0 \Rightarrow$ Let $\lambda = \omega^2$

Case 2 : $\lambda = 0$

$\boxed{\text{Case 1, } \lambda > 0}$

Solving for (3)

$\therefore$ General soln for $T(\theta)$ in case 1

$$T(\theta) = A\cos(\omega\theta) + B\sin(\omega\theta)$$

Since $T(\theta) = T(\theta + 2\pi)$

$$\Rightarrow A\cos(\omega\theta) + B\sin(\omega\theta)$$

$$= A\cos(\omega\theta + \omega 2\pi) + B\sin(\omega\theta + \omega 2\pi)$$

For this to be true

$\Rightarrow \omega = n$, where $n = 1, 2, 3 \ldots$

$\therefore$ General soln for $T(\theta)$ in case 1

$$\boxed{T_{n_1}(\theta) = A_n\cos(n\theta) + B_n\sin(n\theta)}$$

Solving for (2)

Since $\omega = n \Rightarrow \lambda = n^2$

$$\Rightarrow r^2R'' + rR' - n^2R = 0$$

This is a Cauchy-Euler Eqn,

$\Rightarrow$ Guess : $R(r) = Cr^m$

$\vdots$

$m = \pm n$ [Real Distinct]

$\Rightarrow$ General soln for $R(r)$ for case 1

$$R_{n_1}(r) = C_1 r^n + C_2 r^{-n}$$

[Continue Next Pg]

impossible

Centre of disc $u(r,\theta) \to \infty$

As $r \to 0$, $r^{-n} \to \infty$, $R(r) \to \infty$

$\therefore$ We set $C_2 = 0$

$\Longrightarrow \boxed{R_{n_1}(r) = C_1 r^n}$

$\therefore$ For case 1, the general solution is

$u_{n_1} = R_{n_1}(r) \cdot T_{n_1}(\theta)$

$= C_1 r^n [A_n \cos(n\theta) + B_n \sin(n\theta)]$

$u_{n_1} = r^n [a_n \cos(n\theta) + b_n \sin(n\theta)]$

$\boxed{\text{Case } 2,\ \lambda = 0}$

Solving for ③

$T'' + 0(T) = 0$

$T'' = 0$

$T' = A$

$T = A\theta + B$

Since $T(\theta) = T(\theta + 2\pi)$

$T(0) = T(2\pi)$

$A(0) + B = A(2\pi) + B$

$A = 0$

$\therefore \boxed{T_2(\theta) = B}$

Solving for ②

$r^2 R'' + rR' = 0$

$\frac{d}{dr}(rR') = 0$

$rR' = C_1$

$R' = \frac{C_1}{r}$

$R = C_1 \ln(r) + C_2$

impossible

$u(r,\theta) \to -\infty$

As $r \to 0$, $\ln(r) \to -\infty$, $R(r) \to -\infty$

$\therefore$ we set $C_1 = 0$

$\Longrightarrow \boxed{R_2(r) = C_2}$

$\therefore$ For case 2, the general solution is

$u_2 = R_2(r) \cdot T_2(\theta)$

$u_2 = B \cdot C_2$

$u_2 = a_0$

$\therefore$ By using superposition,

$u(r,\theta) =$ General soln of case 2 + General soln of case 1

$u(r,\theta) = a_0 + \sum_{n=1}^{\infty} r^n [a_n \cos(n\theta) + b_n \sin(n\theta)]$

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By using boundary values, $u(1,\theta) = \sin\theta$

$$\Rightarrow a_0 + \sum_{n=1}^{\infty} 1^n [a_n \cos(n\theta) + b_n \sin(n\theta)] = \sin\theta$$

$$\Rightarrow a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\theta) + b_n \sin(n\theta)] = 0 + 1\sin(1\theta)$$

No cos on RHS

By comparison $\Rightarrow a_0 = 0$, $a_n = 0$

$\Rightarrow b_n \sin(n\theta) = 1 \sin(1\theta)$

$\Rightarrow \boxed{n = 1,\ b_1 = 1}$ This mean $b_{n\neq 1} = 0$

$$\therefore u(r,\theta) = 0 + \sum_{n=1}^{\infty} r^{(1)} [0 + [b_1 \sin(\theta) + 0]]$$

$$u(r,\theta) = r\sin(\theta)$$ #

Example 2

Example 6.5.1

Find the solution of Laplace's equation on a half disc of radius c units

$$\{ (r,\theta) \in \mathbb{R}^2 : \ 0 \le r \le c, \ 0 \le \theta \le \pi \}$$

with the boundary conditions

$$u(r,0) = 0 \ \text{ and } \ u(r,\pi) = 0 \ \text{ for } 0 \le r \le c$$

and

$$u(c,\theta) = u_0 \ \text{ for } 0 \le \theta \le \pi$$

where $u_0 > 0$ is a constant.

Given the general solution : $u(r,\theta) = A_0 + \sum_{n=1}^{\infty} r^n [A_n \cos(n\theta) + B_n \sin(n\theta)]$

Using $u(r,0) = 0$

$$A_0 + \sum_{n=1}^{\infty} A_n r^n = 0$$

By comparison $\Rightarrow \boxed{A_0 = 0, \ A_n = 0}$

$$\therefore u(r,\theta) = \sum_{n=1}^{\infty} B_n r^n \sin(n\theta)$$

Using $u(r,\pi) = 0$

$$u(r,\pi) = \sum_{n=1}^{\infty} B_n r^n \sin(n\pi)$$

$$0 = 0$$

Using $u(c,\theta) = u_0$

$$\sum_{n=1}^{\infty} B_n c^n \sin(n\theta) = u_0$$

Using Fourier Formula [Half-Range Expansion]

$$\Rightarrow B_n c^n = \frac{2}{\pi} \int_0^{\pi} u_0 \sin(n\theta) \, d\theta$$

$$= \frac{2u_0}{\pi} \left[-\frac{1}{n} \cos(n\theta) \right]_0^{\pi}$$

$$= \frac{2u_0}{n\pi} [1 - (-1)^n]$$

When n even $\Rightarrow [1-1] = 0$

When n odd $\Rightarrow [1-(-1)] = 2$

$$\therefore B_n c^n = \frac{2u_0}{n\pi}(2) = \frac{4u_0}{n\pi}$$

$$\boxed{B_n = \frac{4u_0}{n\pi c^n} \quad \text{for } n = 1, 3, 5, \ldots}$$

$$\therefore u(r,\theta) = \sum_{n=1,3,5\ldots}^{\infty} \left[r^n \left(\frac{4u_0}{n\pi c^n} \right) \sin(n\theta) \right]$$

$$u(r,\theta) = \frac{4u_0}{\pi} \sum_{n=1,3,5\ldots}^{\infty} \frac{1}{n} \left(\frac{r}{c} \right)^n \sin(n\theta) \quad \#$$